

1 Three Dimensional Coordinate Systems and Vectors

1.3 The Dot Product

Definition 1.3.1 (The Dot Product) If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then the dot product $\vec{a} \cdot \vec{b}$ is given by

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

Example 1.3.2 if $\vec{a} = \langle 1, 0, 3 \rangle$ and $\vec{b} = \langle 2, -1, 1 \rangle$, compute $\vec{a} \cdot \vec{b}$.

Theorem 1.3.3 (Properties of the Dot Product) if $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then

1. $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
2. $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
3. $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
4. $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$
5. $\vec{0} \cdot \vec{a} = 0$

Theorem 1.3.4 If θ is the angle between vectors $\vec{a}$ and $\vec{b}$, then

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta.$$

Example 1.3.5 Find the angle between the vectors $\vec{a} = \langle 2, 2, -1 \rangle$ and $\vec{b} = \langle 5, -3, 2 \rangle$.

Theorem 1.3.6 Two vectors are orthogonal (perpendicular) if their dot product is zero.

1.3.1 Direction Angles

Definition 1.3.7 The direction angles of a nonzero vector $\vec{a}$ are the angles α , β , γ (all in the interval $[0, \pi]$) that $\vec{a}$ makes with the positive x , y , and z -axes, respectively.

Example 1.3.8 Find the direction angles of the vector $\vec{a} = \langle 1, 2, 3 \rangle$.

1.3.2 Projections

We will often need to break vectors up into components to determine how much force or work is being done in a certain direction. More specifically, if we are given two vectors $\vec{a}$ and $\vec{b}$, we wish to determine the projection of $\vec{a}$ onto $\vec{b}$.

Example 1.3.9 Find the projection of the vector $\vec{a} = \langle 1, 1, 2 \rangle$ onto the vector $\vec{b} = \langle -2, 3, 1 \rangle$.

Example 1.3.10 One use of projections is to find the amount of work done by a constant force. A wagon is being pulled a distance of 100 m along by a constant force of 70 N. The handle of the wagon is held at an angle of 35° above the horizontal. Find the work done by the force.